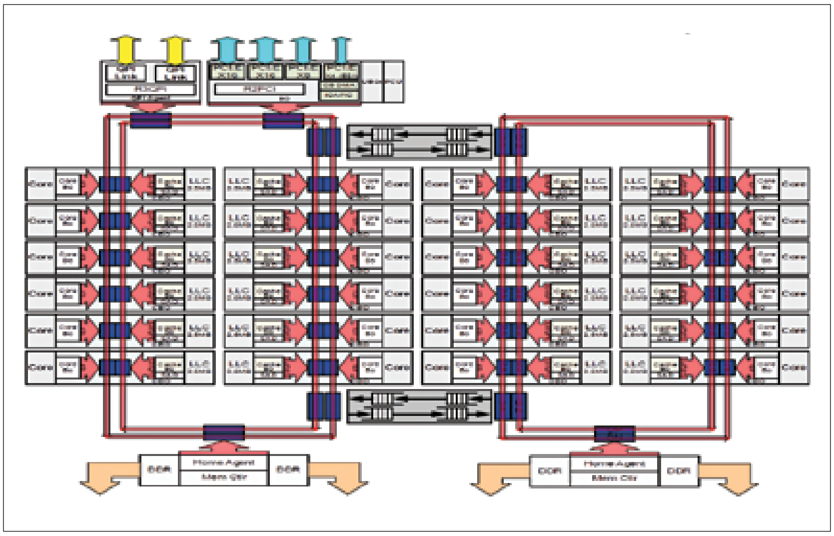
Next Gen Intel
Intel has launched a new core i9 model with 18 cores and 36 threads in addition to the core X upgrades for i3 and i5 **http://dgit.in/CoreXi9**

North Korean
Sensitive information from North Korean defector Thae Yong-ho could be the key to bringing down Kim Jong-un's regime **http://dgit.in/NKoutDe**



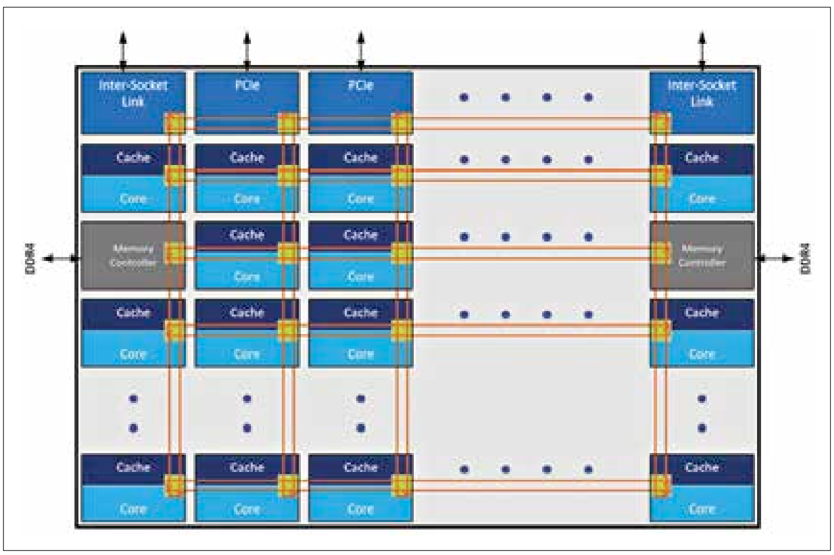
Intel Xeon Processor E5 v4 Product Family HCC

decades and it is currently the most popular method of interconnecting for extremely large number of nodes. It was only natural for a similar system to be implemented for processors.

INTRODUCING THE INTEL MESH ARCHITECTURE

Intel Mesh Architecture first made its foray into the limelight with Knights Landing or Xeon Phi as it's better known. Knights Landing had 72 cores split into two segments of 36 cores each. All of these cores were arranged according to the new Mesh Architecture. And the upcoming Skylake-X and Skylake-SP Xeon processors will be utilising the Intel Mesh Architecture.

While this is just a sectional diagram of the entire processor, it is more than sufficient to get the hang of Mesh Architecture. Ensuring a fast interconnect between cores and reduced latency is what this architecture is all about. With the Ring Bus architecture, adding more cores would have required adding more rings but they'd then take up a certain footprint on the die which could otherwise have been allocated for the cores. And latency would surely have increased, as moving data from one core to another would have required even more clock cycles.



Mesh architecture

With the Mesh Architecture, you can scale in an ever-increasing fashion. In this architecture, all cores are arranged in a grid like manner and each core is connected to its neighbouring cores through a mesh formed by rows and columns of interconnect paths. The number of interconnect paths will scale according to the number of cores present on the die.

Communicating along a column is quicker as compared to communicating along a row but not in all cases. If the core in the bottom-left corner were to communicate with the core right above it or right beside it, then it would take exactly one cycle. Let's call this a unit distance. However, if it has to communicate with a core along the row that's two units away then it will take 3 cycles. Three units away and it will take 3+1 = 4 cycles. And so on. Vertically, it will continue to consume just one cycle per core. Thus, communicating with cores that are farthest away requires a lot more cycles horizontally than vertically.

With Skylake-X and Skylake-SP, Intel's QPI will make way for UPI or UltraPath Interconnect (also known as Keizer Technology Interconnect or KZI). Not much is known about this new interconnect aside from the fact that it is obviously faster than the previous QPI. And traffic will flow in both directions on this bus.

Another key advantage of the Mesh Architecture is the reduced power consumption. Since communication between cores need not necessarily take round trips about a ring bus, it ends up with a far higher amount of intrinsic bandwidth. The net result is lower power consumption for the communication which can then be offset onto the cores for compute purposes. Hence, you can get more performance with the same TDP.

Overall, the new Intel Mesh Architecture has been built from scratch with the intention of scalability for generations to come. With AMD entering the server market with its EPYC 7000 series processors, the race has begun anew. d